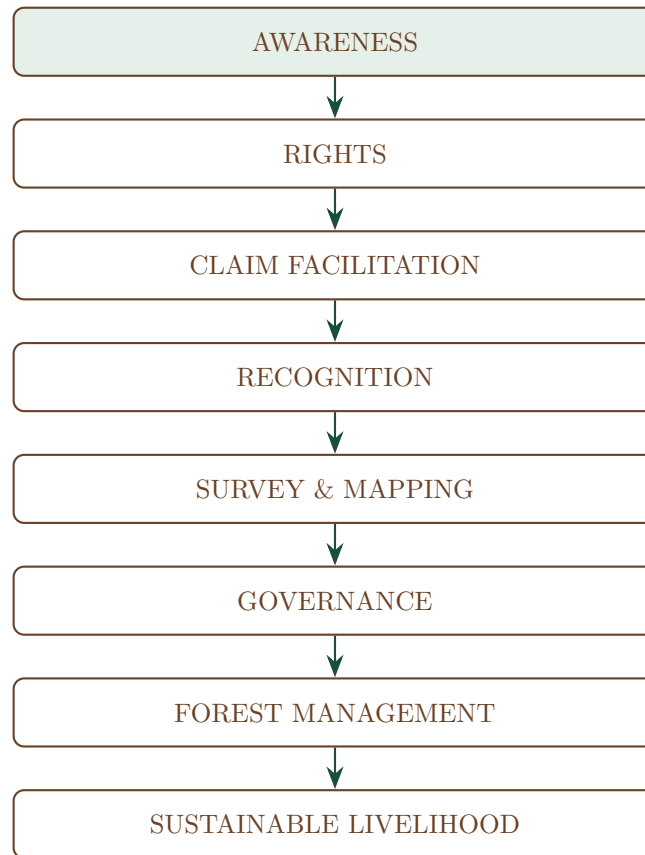


VanMitra

Final Application Architecture

A Community-Centred Digital Ecosystem for FRA, CFR, Land Mapping, Gram Sabha Governance and Sustainable Forest Livelihoods



August 31, 2026

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Chapter 1

Introduction

VanMitra is proposed as an **offline-first, community-centred digital platform** designed to support the on-ground implementation of the **Forest Rights Act (FRA)**, with special emphasis on **Community Forest Rights (CFR)** and community forest governance.

The final architecture combines:

- the original VanMitra AI architecture;
- real-world problems identified during NGO interaction;
- the proposed on-ground implementation scheme;
- the need for coordination with NGOs and government institutions;
- the existing digital FRA ecosystem;
- the problem of pending land surveys and unclear boundaries;
- the need for transparent Gram Sabha governance; and
- the long-term objective of converting recognised CFR into sustainable community management and livelihoods.

The architecture therefore moves beyond a simple **claim-filing application**.

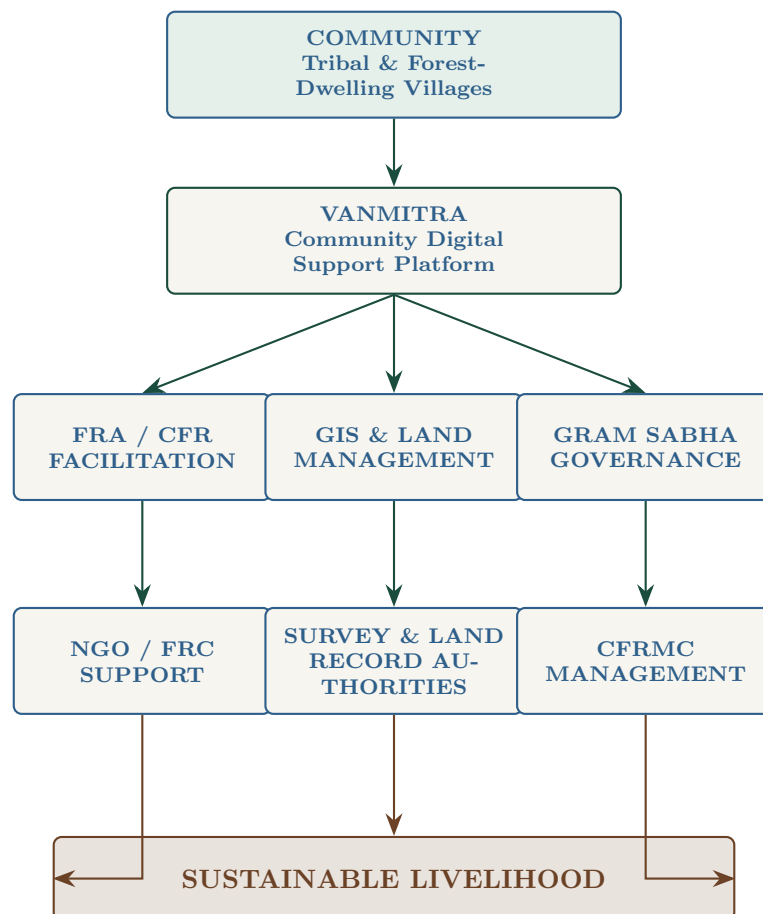
Awareness → Rights → Claim Facilitation → Recognition → Survey & Mapping → Governance → Forest Management → Sustainable Livelihood

The system is designed as a **support and facilitation platform**. It does not replace the legal authority of the Gram Sabha, FRC, statutory committees, land-survey authorities or government departments. This principle is central to the architecture.

Chapter 2

Final Vision of VanMitra

VanMitra functions as an integrated ecosystem connecting five major dimensions.



The platform supports the complete community journey rather than only one stage of the FRA process.

Chapter 3

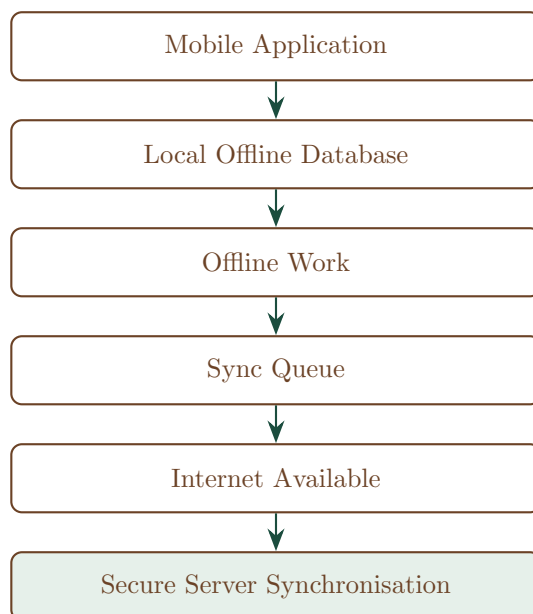
Core Design Principles

3.1 Community First

The system should be understandable and usable by tribal communities, forest dwellers, Gram Sabha members, FRC members, CFRMC members, local volunteers, and NGO field workers. Technology should adapt to the community rather than expecting the community to adapt to complex technology.

3.2 Offline First

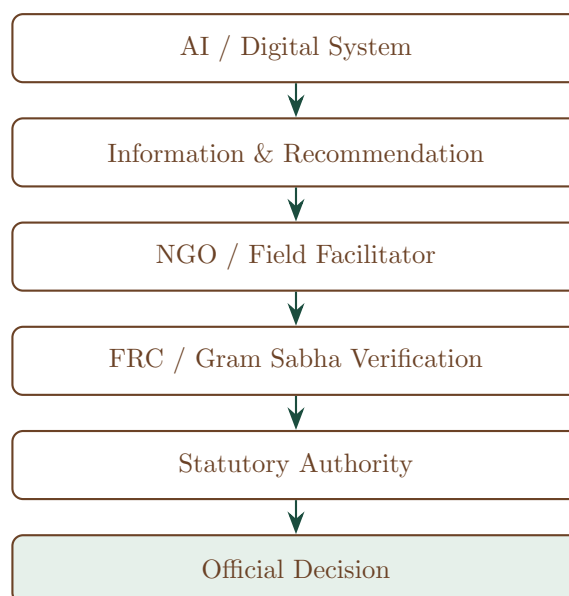
Target villages may have weak internet connectivity, intermittent mobile networks, and limited access to continuous digital infrastructure.



The original VanMitra technical architecture already proposed an offline-capable, mobile-first system using Flutter and local storage with backend synchronisation.

3.3 Human-in-the-Loop

VanMitra must never automatically decide claim eligibility or approval. Instead:



VanMitra assists the process; it does not replace statutory institutions.

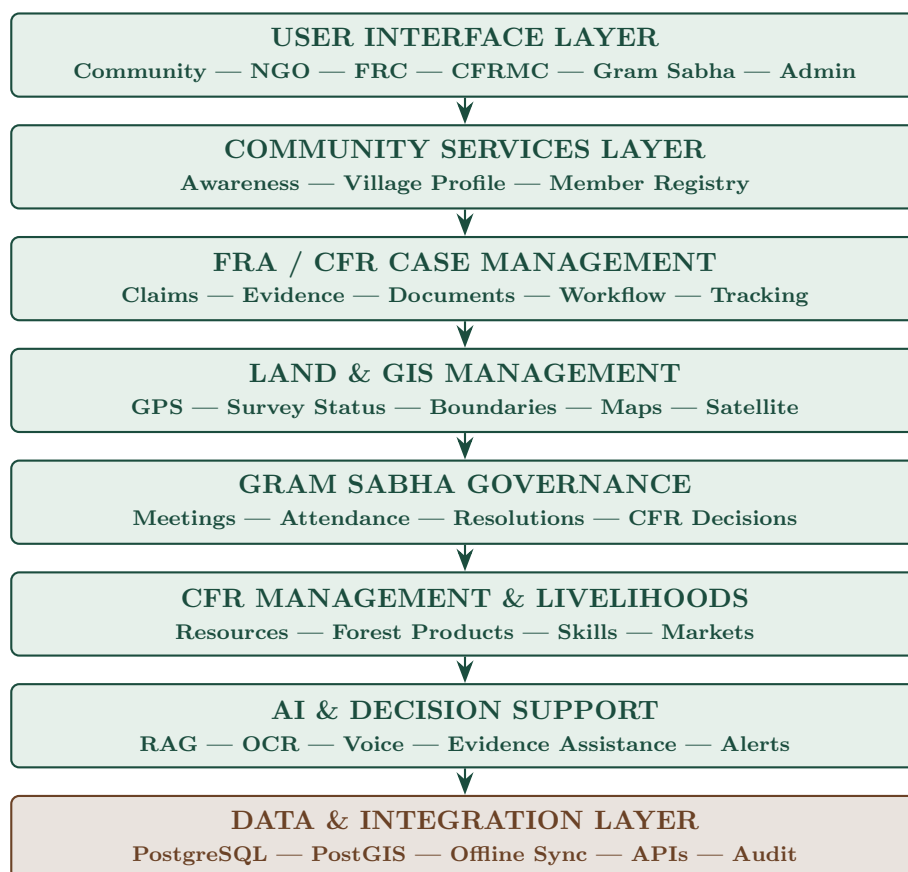
3.4 Government-Compatible, Not Government-Replacing

The platform works around and alongside authorised government processes. VanMitra should not attempt to replace official FRA portals, FRC, Gram Sabha, Sub-Divisional Level Committee, District-Level Committee, official land survey, cadastral records, or statutory decision-making. Instead, VanMitra becomes the **community-side evidence, workflow, GIS and governance layer**.

Chapter 4

Final High-Level Application Architecture

The final VanMitra architecture consists of **eight major functional layers**.



Chapter 5

User and Stakeholder Architecture

The application does not provide the same interface to every user. Different stakeholders require different levels of access.

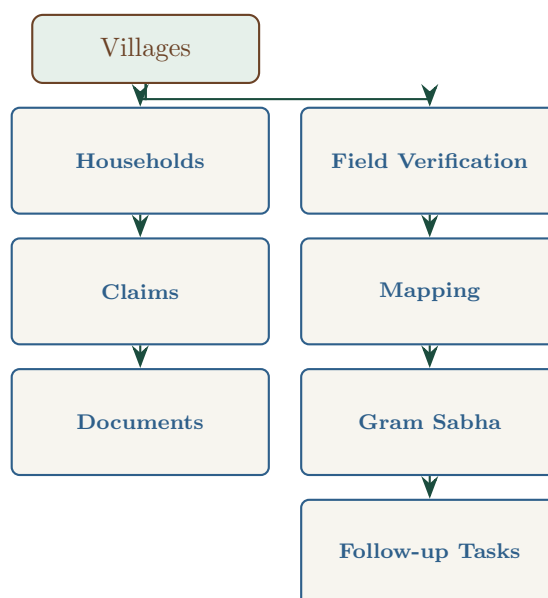
5.1 Tribal Community Member

Main functions: My Village, My Rights Information, My Claim, My Documents, My Land, Gram Sabha, Village Forest, Help & Support.

The community member should be able to understand FRA/CFR, access awareness material, see claim information, upload documents with assistance, view land information, view authorised Gram Sabha records, see upcoming meetings, and understand the next required step.

5.2 NGO Field Worker

The NGO application is one of the most important components of the architecture.



NGO workers can onboard villages, register community cases, collect evidence, upload documents, conduct field verification, capture GPS information, assist participatory mapping, prepare meeting records, track pending cases, and coordinate with relevant institutions.

5.3 FRC Interface

Claim list; evidence checklist; verification records; village-level documentation; claim readiness; Gram Sabha linkage; pending actions.

5.4 CFRMC Interface

CFR Area, Forest Resources, Management Activities, Village Decisions, Resource Survey, Livelihood Opportunities, Monitoring Alerts. The CFRMC uses VanMitra as a **community forest management support tool**.

5.5 Gram Sabha Interface

Upcoming meetings; agenda; member attendance; participation statistics; resolutions; village claims; CFR map; important community decisions.

5.6 Government/Administrative Coordination Interface

An initially controlled dashboard, not an attempt to create a parallel government system. May support survey-pending cases, formal application preparation, verified documentation packets, status updates, and institutional coordination. Access is provided only where officially authorised.

Chapter 6

Module 1–5: Awareness to AI-Assisted Support

6.1 Module 1: FRA and CFR Awareness & Community Mobilisation

The entry point of the VanMitra ecosystem. The implementation scheme proposes village-level awareness and community mobilisation involving NGOs, community institutions and, where appropriate, designated government representatives.

FRA Knowledge Centre topics: What is FRA? Individual Forest Rights; Community Rights; Community Forest Rights; Role of Gram Sabha; Role of FRC; CFR process; rights and responsibilities after recognition.

Local Language Support: Marathi, Hindi, English, and local tribal languages where technically and linguistically possible.

Multimedia formats: Text, Audio, Illustrations, Short Videos, Voice Guidance, Interactive Process Diagrams.

Village Awareness Dashboard records: awareness sessions conducted, number of participants, women participants, households covered, questions raised, potential eligible communities identified.

6.2 Module 2: Village and Community Registry

Village Profile fields: Village ID, Village Name, Gram Panchayat, Taluka, District, State, Population, Number of Households, Connectivity Status, Forest Dependency Profile, Existing FRA Status, CFR Status.

Community Institution Registry: Gram Sabha, FRC, CFRMC, community representatives, authorised NGO facilitators.

6.3 Module 3: FRA/CFR Digital Case Management

The central operational module. Every case has a unique digital identity, organised as IFR Case, Community Rights Case, or CFR Case under each Village. Each case contains: Community/Claimant Information, Claim Type, Documents, Evidence, Land/Forest Information,

Field Verification, Gram Sabha Record, Submission Information, Institutional Status, Survey Status, and Final Outcome.

6.4 Module 4: Evidence and Document Management

Prevents documents from being scattered across paper files, NGO offices, mobile phones, WhatsApp, and individual volunteers.

Evidence Categories: Identity Documents → Residence Documents → Community Information → Historical Evidence → Land/Forest Evidence → Field Evidence → Gram Sabha Documents.

Document Metadata: Document ID, Case ID, Document Type, Upload Date, Uploaded By, Verification Status, Verified By, Remarks.

6.5 Module 5: AI-Assisted Claim and Document Support

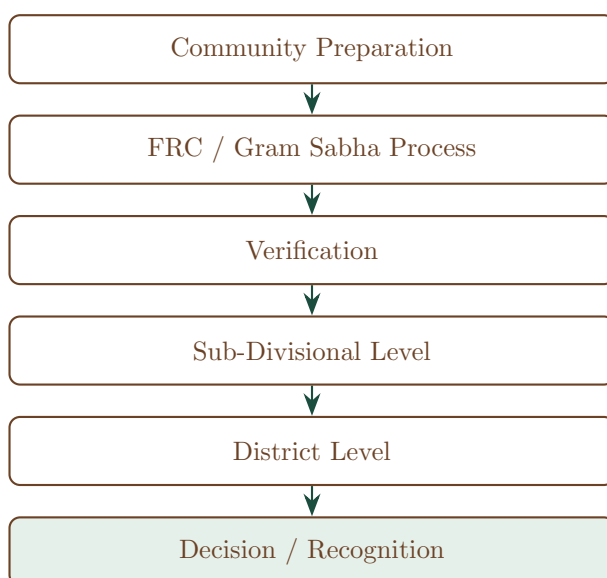
AI is a **supporting service**, not the centre of the application.

- **Document Assistance:** OCR for scanned documents, document classification, extraction of important fields, duplicate detection.
- **Voice Assistance:** Community Member Speaks → Speech-to-Text → Structured Information → Field Worker Verification.
- **Knowledge Assistant:** Answers what the next process is, which documents are missing, what a status means, and what should be prepared next — based on a verified knowledge base.
- **Claim Readiness Indicator:** An *administrative* readiness indicator (Documents, Evidence, Field Verification, Gram Sabha completeness) — **not** a legal approval score.

Chapter 7

Module 6–11: Workflow, Survey and GIS

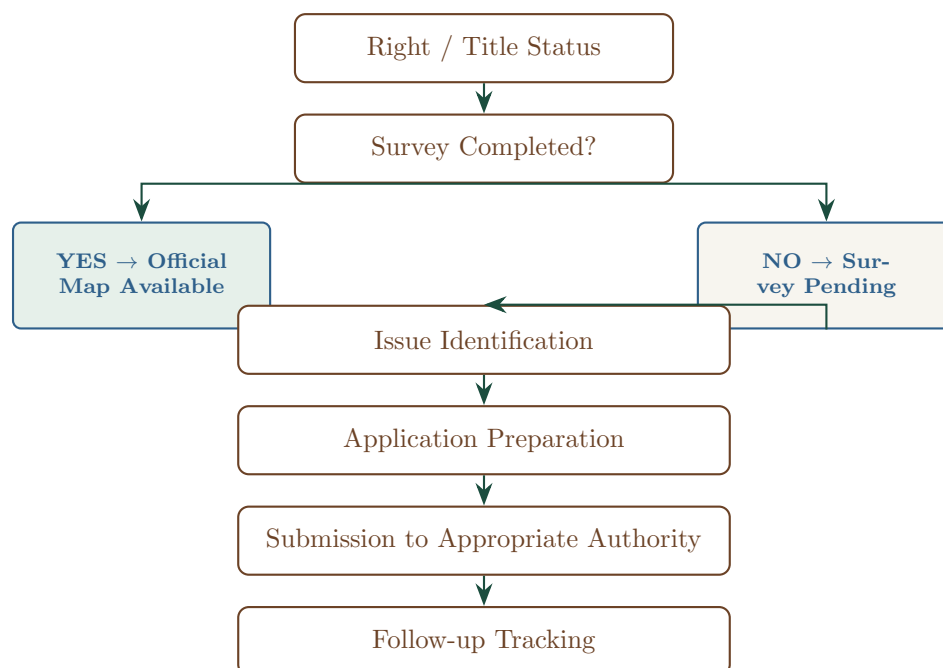
7.1 Module 6: Step-by-Step Institutional Workflow Tracker



For each case, users can see current stage, previous stage, submission date, pending action, documents submitted, and next expected action.

7.2 Module 7: Land Survey and Boundary Coordination

A right may be recognised while the community still lacks proper measurement, clear boundary information, or usable survey records. VanMitra contains a dedicated **Survey Status Management System**.



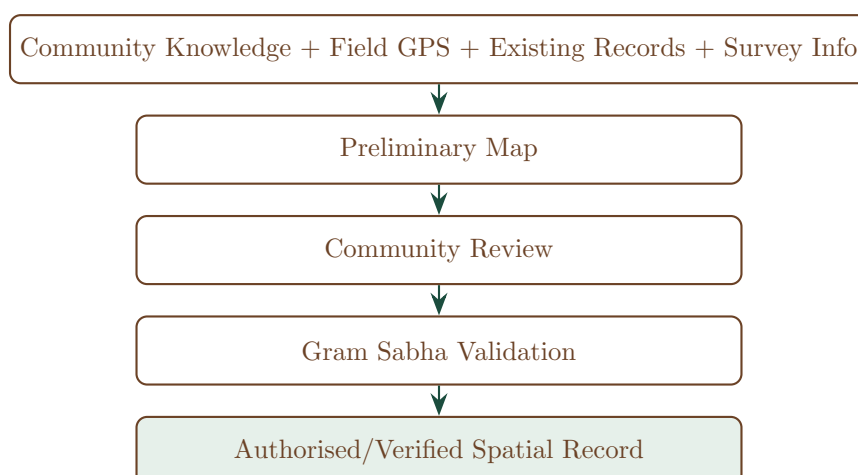
The implementation plan identifies coordination with land-record and survey institutions such as Maharashtra’s relevant land-record authorities, where applicable.

7.3 Module 8: GIS and Geospatial Land Management

GIS Data Layers (Base Map): Village Boundary, Forest Area, CFR Boundary, Community Use Areas, Water Resources, Paths and Access Routes, Natural Resources, Traditional Use Areas, Community Management Zones.

The implementation includes integration of verified spatial information, community field information and Gram Sabha validation into a digital geospatial database.

7.4 Module 9: Participatory Mapping



NGO workers capture GPS points, walking tracks, landmarks, photographs, water sources, forest-use areas, and traditional locations.

7.5 Module 10: Official Survey vs Community Mapping

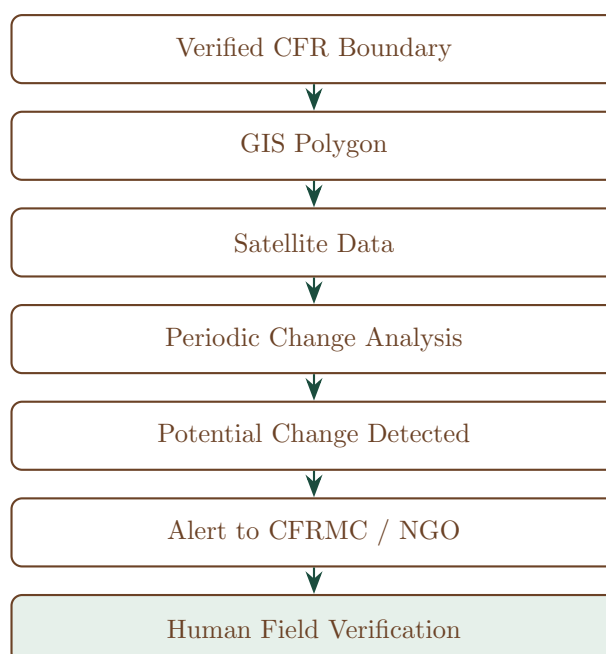
Every boundary carries a classification:

Status	Meaning
● Officially Surveyed	Legally authoritative cadastral survey completed
● Verified Spatial Record Available	Spatial record exists and has been verified
● Community/Participatory Mapping	Mapped through community participation, not yet authorised
● Boundary Unclear / Survey Pending	No reliable boundary information available

The implementation document explicitly states that VanMitra’s GPS and mapping tools are for preliminary mapping, participation and survey coordination, and should **not** replace an officially authorised cadastral survey.

7.6 Module 11: Satellite and Forest Change Monitoring

The original VanMitra system included a satellite-based digital fencing concept using Sentinel imagery for monitoring recognised community forest boundaries.



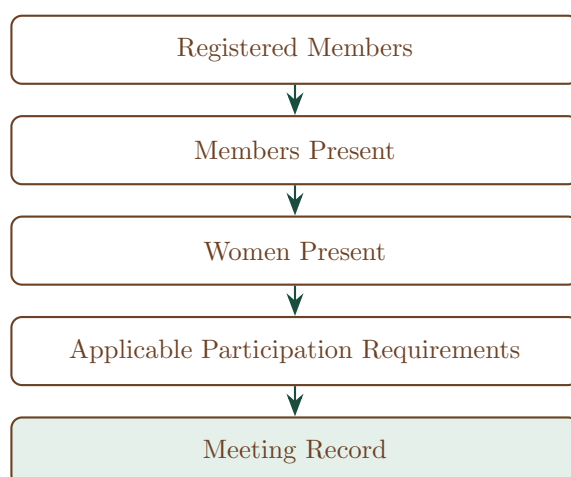
The system never automatically declares “Illegal encroachment detected.” Instead it reports: *“Potential land-use or vegetation change detected. Field verification recommended.”*

Chapter 8

Module 12–16: Gram Sabha Governance and Livelihoods

8.1 Module 12: Gram Sabha Digital Governance

Meeting Creation records meeting date, location, agenda, and purpose.



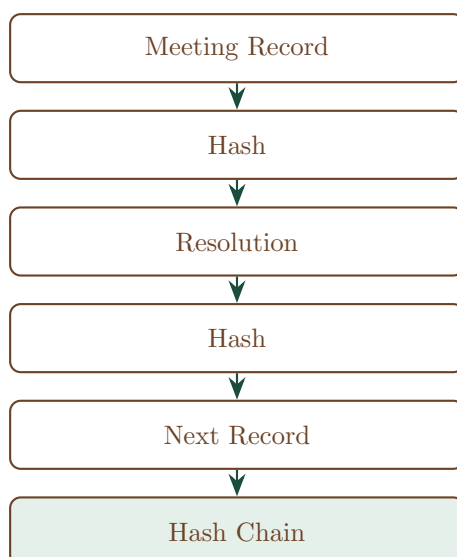
The implementation scheme specifically includes recording participation and women’s inclusion as part of structured meeting documentation.

8.2 Module 13: Digital Resolution Management

Each resolution becomes a structured digital record: Resolution ID, Meeting ID, Date, Agenda, Discussion, Decision, Resolution Text, Supporting Documents, Verification, Historical Record. Resolutions can be linked to a CFR claim, land issue, survey request, forest-management decision, or livelihood activity. The goal is transparency and continuity of records, while continuing to respect official legal procedures.

8.3 Module 14: Tamper-Evident Record System

The original technical architecture proposed a tamper-evident Gram Sabha ledger using hash chaining.



This allows the system to detect whether stored records have been modified without updating the authorised history. A full blockchain is **not** necessary for the initial pilot.

8.4 Module 15: CFR Resource Management

After recognition, VanMitra transitions from **rights facilitation** to **community forest management**. The CFRMC dashboard covers: CFR Area, Resources, Forest Products, Seasonal Calendar, Management Activities, Community Decisions, Monitoring, Livelihood Opportunities.

8.5 Module 16: Village Resource and Livelihood Assessment

The implementation scheme proposes a Village Resource and Livelihood Profile containing: available resources, seasonal availability, community skills, potential products, market connections, value-addition opportunities, and sustainable utilisation plans.

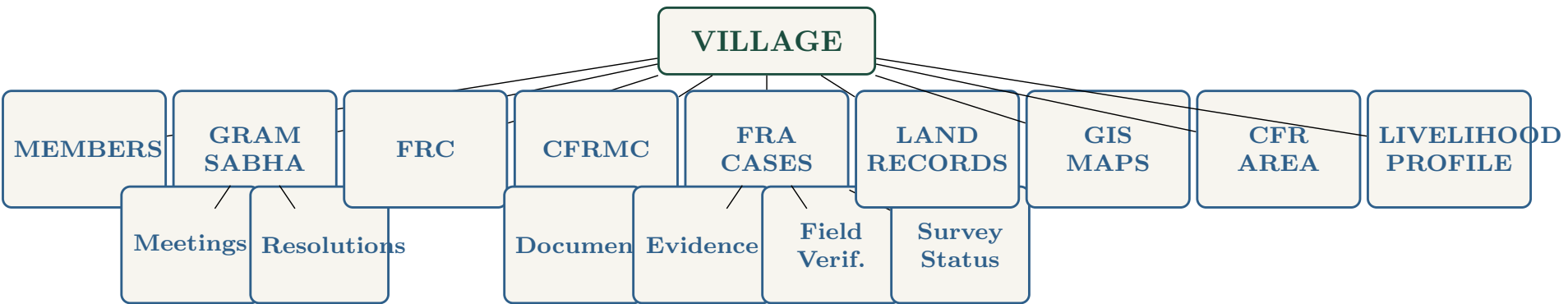


Module 17: Village Opportunity Engine (Future Component)

Analyses Forest Resources + Community Skills + Seasonality + Existing Products + Market Information to generate *suggestions*: possible value-added products, training requirements, resource-management opportunities, and potential cooperative activities. These remain recommendations for discussion by the community, CFRMC, Gram Sabha, NGOs, and researchers.

Chapter 9

Final Data Architecture



Chapter 10

Final Technical Architecture

10.1 Frontend — Mobile Application (Flutter)

Used for Android devices, offline functionality, multilingual interface, field data collection, GPS integration, and camera/document capture.

10.2 Local Storage

Hive / SQLite — stores cases, documents, pending forms, GPS points, meeting data, and the sync queue.

10.3 Backend (FastAPI)

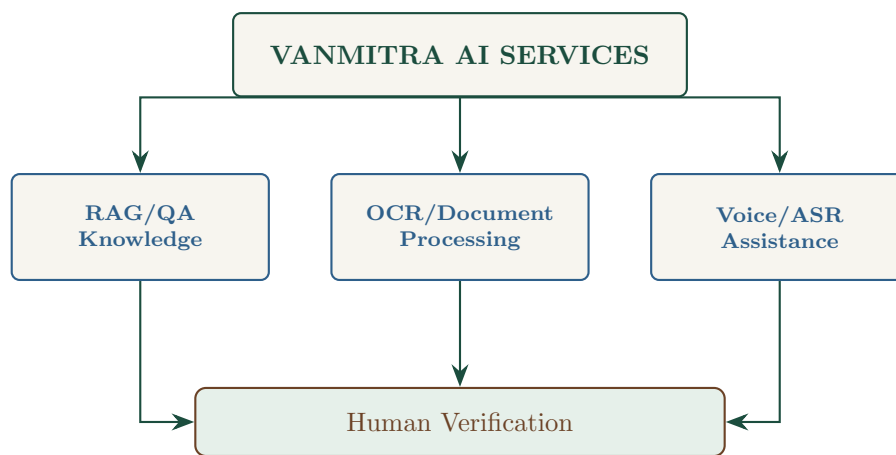
Responsible for APIs, authentication, case management, workflow management, AI integration, and data synchronisation.

10.4 Database

Component	Purpose
PostgreSQL	Users, cases, documents, meetings, workflows
PostGIS	Polygons, GPS points, village boundaries, CFR areas, spatial analysis

Chapter 11

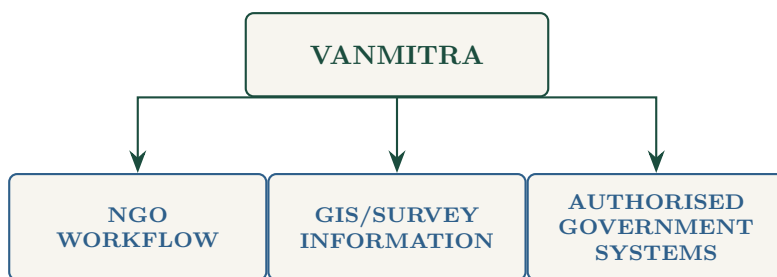
Final AI Architecture



AI remains an assistant. It should not replace legal verification, community decision-making, official land survey, or statutory approval.

Chapter 12

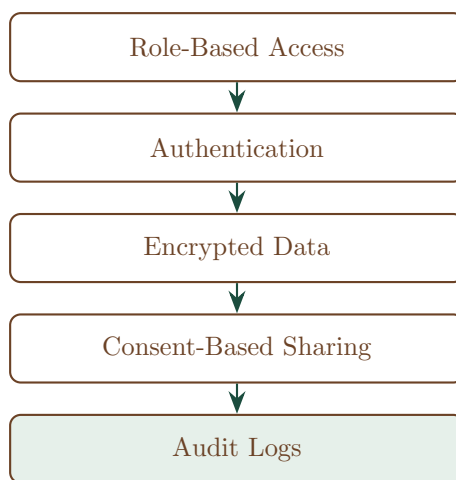
Integration Architecture



Potential future integrations should only happen through authorised APIs, official data-sharing mechanisms, and formal institutional permissions.

Chapter 13

Security and Privacy Architecture



User	Access
Community Member	Personal/community authorised data
NGO Worker	Assigned village/cases
FRC	Relevant claims
CFRMC	CFR management data
Gram Sabha	Authorised community records
Admin	System administration

Chapter 14

Final End-to-End VanMitra Workflow

Phase	Description
1	Awareness
2	Village & Community Onboarding
3	Identification of Potential Rights
4	Claim / CFR Case Creation
5	Document & Evidence Collection
6	Field Verification & Participatory Mapping
7	Gram Sabha Discussion & Resolution
8	Statutory Process Tracking
9	Right Recognition
10	Survey Status Check
11	Survey Coordination if Pending
12	Geospatial CFR Map
13	CFRMC Forest Management
14	Resource & Livelihood Development

This directly reflects the proposed implementation scheme, which extends from awareness and claim preparation to survey coordination, geospatial mapping, Gram Sabha governance and sustainable livelihood development.

Chapter 15

Final Application Navigation Architecture

- **Home**
- **FRA & CFR Awareness**
- **My Village**
- **Claims & Cases**
 - Documents
 - Evidence
 - Field Verification
 - Status
- **My Land / CFR Map**
 - Boundary
 - Survey Status
 - Forest Monitoring
- **Gram Sabha**
 - Meetings
 - Attendance
 - Resolutions
- **CFR Management**
 - Resources
 - Activities
 - Monitoring
- **Livelihood Opportunities**
- **Help & NGO Support**

Chapter 16

Final Architecture Summary

The final VanMitra application is not simply an **AI claim-filing system**. It is a **multi-stakeholder, community governance and forest-rights ecosystem**.

1. FRA/CFR Awareness Module
2. Village & Community Registry
3. FRA/CFR Digital Case Management
4. Evidence & Document Management
5. AI-Assisted Documentation Support
6. Step-by-Step Institutional Tracking
7. Land Survey & Boundary Coordination
8. GIS and Geospatial CFR Mapping
9. Participatory Community Mapping
10. Satellite Change Monitoring
11. Gram Sabha Meeting Management
12. Digital Resolution Management
13. Tamper-Evident Governance Records
14. CFRMC Forest Management
15. Village Resource Assessment
16. Community Livelihood Development
17. NGO Field Implementation Dashboard
18. Government/Institutional Coordination Layer
19. Offline Synchronisation Architecture
20. Privacy, Security and Role-Based Access

Final Architectural Vision



VanMitra's Final Purpose

VanMitra will help tribal and forest-dwelling communities move from awareness of their forest rights to the documentation, recognition, mapping, governance, management and sustainable utilisation of those rights.

Most importantly: VanMitra does not decide who receives forest rights. It empowers communities and supporting institutions with better information, documentation, mapping, transparency and coordination throughout the legal and community governance process.